Hologram counts

How Holograms Work

Name _____ Date _____

Show your thinking with numbers, marks, or words.

1. A sheet has 62 shiny squares, and 37 of them hold a bird. How many squares have no bird?

62 − 37 = _____

Show your thinking.

2. Little Wonderer counts 27 ripple bumps on one side of the pan and 36 on the other. How many bumps in all?

27 + 36 = _____

Show your thinking.

3. A strip of film shows 54 stripes, and 28 are dark. How many stripes are bright?

54 − 28 = _____

Show your thinking.

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4. A hologram plate is 90 millimetres wide. One broken piece is 47 millimetres wide. How wide is the other piece?

90 − 47 = _____

Show your thinking.

5. Big Wonderer tilts the card 19 times, and the little one tilts it 26 times. How many tilts in all?

19 + 26 = _____

Show your thinking.

6. The scientist waits 45 seconds for the table to go still, then records for 38 seconds. How many seconds in all?

45 + 38 = _____

Show your thinking.

WONDER DAILY

How Holograms Work

A bird inside a card

A flat shiny square holds a tiny bird. Tilt the card, and you can see around its sides.

Two ripples cross

Two ripple rings cross in a pan. Bumps stack up, and a bump plus a dip cancels to calm.

Stripes, not a picture

A laser beam splits in two. Where the beams cross on the film, it records stripes, not a picture.

Two halves, both whole

Break a hologram in half. Each half still shows the whole shell, because every spot saw it all.

A rainbow bird

The bird is not painted. Tiny grooves bend light back into the bird, so each eye sees it a new way.

A bird that floats?

Most floating stage pictures are a reflection in slanted glass. Could light ever draw in empty air?

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

Explain a light clue in your words

How Holograms Work · Curiosity day

Name _____ Date _____

Choose one picture. Say your idea first, then draw or write it.



Two ripples cross



Stripes, not a picture



Two halves, both whole

Draw two ripple rings crossing. Mark a spot where they stack up and a spot where they go calm.

wave · ripple · stripe · light · because · maybe

The water goes calm in some spots because … what happens when a bump meets a dip?

The film does not look like a shell because … what does the film record?

Each half shows the whole shell because … what did every part of the film see?

Does your sentence tell what you noticed and use because or maybe to explain?



The Bird Who Lives in a Card

Little Wonderer found a small bird in a card,
not painted, not printed, not stuck on too hard.
Tip it this way, and you'll see its left wing.
Tip it that way, and its right! What a thing!

"It's ripples," the big one said, filling a pan.
"Tap twice in the water as fast as you can.
Where bumps meet with bumps, they jump up with glee;
where bumps meet with dips, they go flat as can be."

The film doesn't hold any bird, any beak,
just stripes upon stripes far too tiny to peek.
Snap it in two, and here's what you'll find:
each half keeps the whole bird. (The bird doesn't mind.)

So Big Wonderer whispered, "Stand still as a stone,
for one wiggle will smudge every stripe that we own."
The little one froze, not a sneeze, not a sound,
and a bird made of light turned slowly around.

How Holograms Work

01 What do ripples have to do with light?



Where ripples cross, bumps stack up or calm out.

Look closer. Find the tall bumps and the flat calm lines where the two rings cross.

A little explanation. Light travels in waves, a bit like ripples. Where two crests meet, they stack into a taller bump. Where a crest meets a dip, they cancel into calm. A hologram records that crossing pattern, made with light instead of water.

Try this. Put a little water in a baking pan. Tap two fingertips in it at the same time. Look for the calm lines between the bumps.

02 What is really on a hologram's film?



The film holds fine stripes, not a shell.

Look closer. Compare the shell with the fine stripes in the lens. Do they look alike?

A little explanation. Not a picture of the shell. A laser beam is split in two. One part bounces off the shell, and one part shines straight on the film. Where they cross, the film records light and dark stripes far thinner than a hair.

Try this. Look at your fingertip with a magnifying glass. Hologram stripes swirl a bit like that, but they are far tinier and closer together.

03 What happens if a hologram breaks?



Each half still shows the whole shell.

Look closer. Look inside each half. Is any part of the shell missing?

A little explanation. Each half still shows the whole shell, just a little blurrier. Light from every spot on the shell landed on every part of the film. It is like peeking through one pane of a window: you still see the whole yard.

Try this. Look out a window through a paper tube. Move the tube to a different pane. Can you still see the whole tree from each spot?

04 Are floating pictures in movies holograms?



Slanted glass makes a reflection seem to float.

Look closer. Find the real toy bird below and its reflection in the slanted glass.

A little explanation. Usually not. Many stage holograms are an old trick from 1862 called Pepper's ghost. A slanted sheet of clear glass reflects a lit figure, so it seems to float. A real hologram needs film or foil to bend the light.

Try this. At night, look out a dark window from a bright room. Can you see your room floating outside? That is the same glass trick.